# Quick Sort

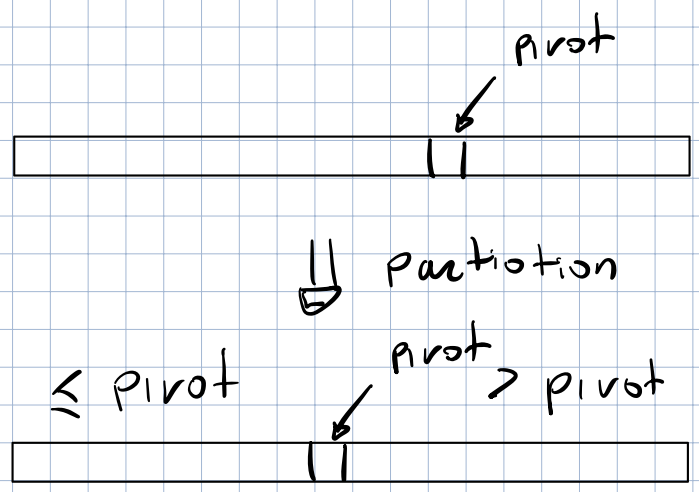
Cormen 7.1, 7.2, 7.3

Another sorting algo based on

- D&C approach
- comparisons

**Merge Sort**
- Divide : Constant Time
- Combine : Linear Time

**Quick Sort**
- Divide : Linear Time
- Combine : Constant Time




5 6 3 2 $\boxed{5}$ 8 4 2

$\downarrow$

$\underbrace{3\ 2\ 4\ 2\ 5}_{QS}\ |\ 5\ |\ \underbrace{6\ 8}_{QS}$

**Divide** Choose a pivot (at random)

Partition the array $A[p, \dots, r]$ in two (+1) (possibly empty) subarrays $A[p, \dots, q-1]$ and $A[q+1, \dots, r]$ such that

- $\forall \ell \in [p \dots q-1]$, $A[\ell] \leq A[q]$
- $A[q]$ = pivot
- $\forall \ell \in [q+1, \dots, r]$, $A[q] < A[\ell]$

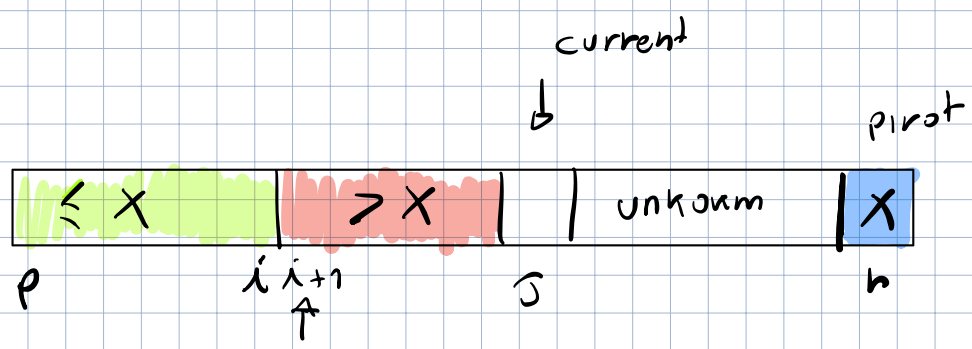
**Conquer** Stop if $p \geq r$

Otherwise sort recursively the two subarrays

**Combine** do nothing

```
QuickSort(A, p, r)
    if p < r
        q = Partition(A, p, r)
        QuickSort(A, p, q-1)
        QuickSort(A, q+1, r)
```

```
Partition(A, p, r)
    // Choose pivot and move it in A[r]
    x = A[r]
    i = p - 1
    for j = p to r-1:        // range(p, r
        if A[j] ≤ x
            i = i + 1
            swap(A[i], A[j])
    swap(A[i+1], A[r])
    return i+1
```




inplace

not stable

Counterexample for stability

1 1 2 → 2 1 1 but easy to fix

pivot

pivot

2 2 1 1 → 1 2 2 1

i j x i

```
Partition (A, p, r)
    // Choose pivot and move it in A[r]
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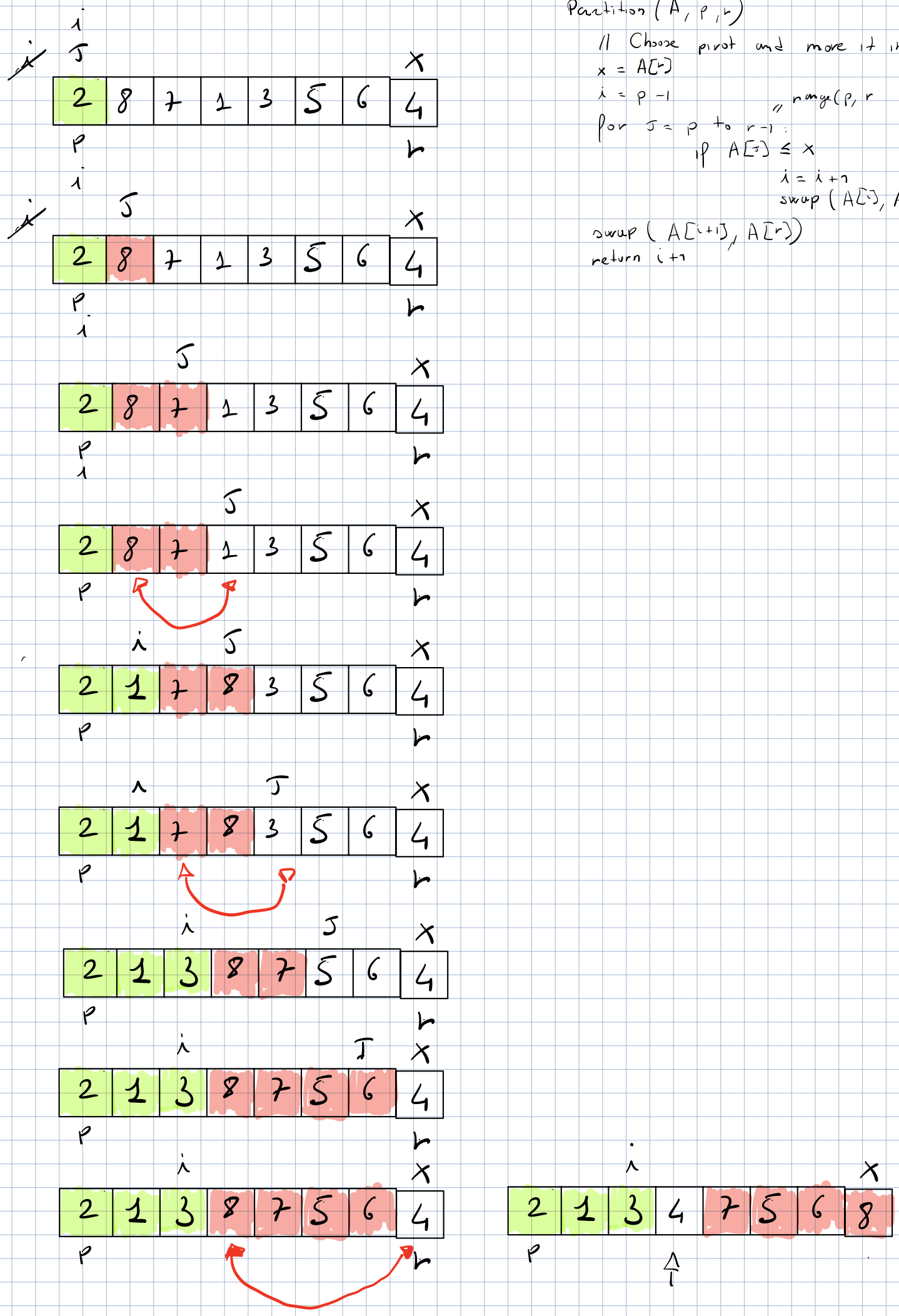


QuickSort(A, p, r)

if $p < r$

$q = \text{Partition}(A, p, r)$

QuickSort(A, p, q-1)

QuickSort(A, q+1, r)

Best Case: Every time we choose the median as the pivot of $A[p \ldots r]$.

$$T(n) = \begin{cases} O(1) & \text{if } n \leq 1 \\ 2\,T\left(\frac{n}{2}\right) + \Theta(n) \end{cases}$$

$\Theta(n \log n)$

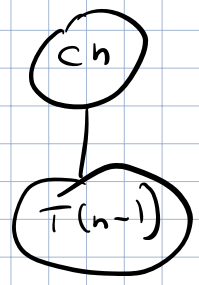
Worst case: Every time we choose the min or max in $A[p \ldots r]$

$$T(n) = \begin{cases} O(1) & \text{if } n \leq 1 \\ T(n-1) + \Theta(n) \end{cases}$$

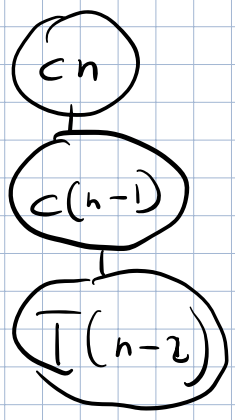
"Same" as Selection Sort so $\Theta(n^2)$

1)

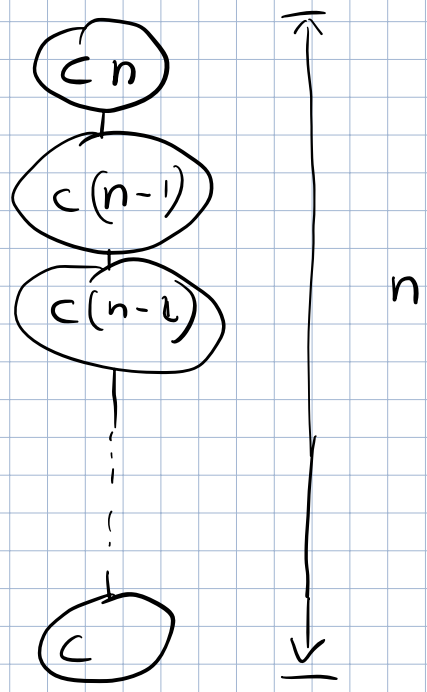
2)




3)



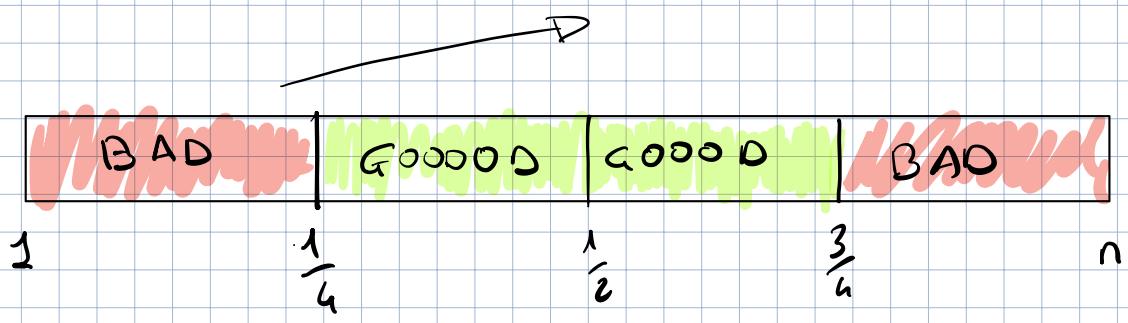

4 - +∞)



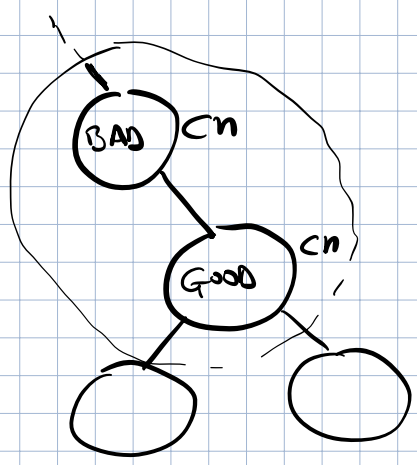

$$c\sum_{i=1}^{n} i = \Theta(n^2)$$

# Average Case Analysis (only intuition)

$$\boxed{\Theta(n \log n) \quad \text{whp} \qquad \text{prob} \quad 1 - \frac{1}{n^2}}$$




$$P(\text{GOOD}) = \frac{1}{2} = P(\text{BAD})$$




| Cost | when | prob | so |
|---|---|---|---|
| $cn$ | GOOD | $\frac{1}{2}$ | |
| $2cn$ | B - G | $\frac{1}{4}$ | $\frac{3}{4}$ |
| $3cn$ | B - B - G | $\frac{1}{8}$ | $\frac{7}{8}$ |
| $4cn$ | B - B - B - G | $\frac{1}{16}$ | $\frac{15}{16}$ |

# k distinct elements

Given an array A of n element but just k of them are distinct.

A 5 6 3 5 3 6 4 3 6 5 $\quad n = 10$

$k = 4$

| < pivot | = pivot | > pivot |
|---|---|---|
| | | |

5 6 5 6 6 6 5 5 6 6 6

$\downarrow$ pivot = 5

$$< \text{pivot} \quad \underbrace{5\,5\,5\,5}_{= \text{pivot}} \; \underbrace{6\,6\,6\,6\,6\,6\,6}_{> \text{pivot}} \searrow \text{QS}$$

$$< \text{pivot} \quad \underbrace{6\,6\,6\,6\,6\,6\,6}_{= \text{pivot}} \quad > \text{pivot}$$

$\Theta(n \log k)$

# Asymptotic Notation $\Theta$

- forget about constant factors and lower order terms

$$100n + 1000 = \Theta(n) \qquad \underline{O(1)}$$

$$\frac{1}{1000} n^2 + 7 = \Theta(n^2)$$

## $\Theta$ - notation

For a given function $g(n)$,

$\Theta(g(n))$ denotes a set of function

$$\Theta(g(n)) = \{ f(n) : \text{there exist positive constants } c_1, c_2, \text{ and } n_0 \text{ such that}$$

$$0 \leq c_1 g(n) \leq f(n) \leq c_2 g(n)$$

$$\text{for all } n \geq n_0 \}$$

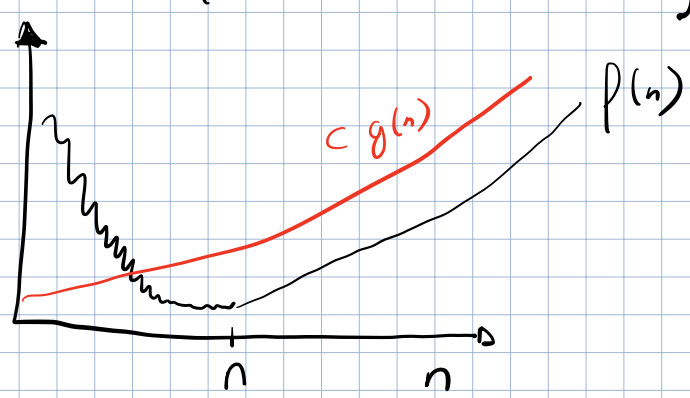
$$100n^2 + 1000n \overset{=}{\cancel{\in}} \Theta(n^2)$$

O - notation  Big-O

For a given function $g(n)$,

$O(g(n))$ denotes a set of function

$$O(g(n)) = \{ f(n) : \text{there exist positive constants } c, \text{ and } n_0 \text{ such that } 0 \leq f(n) \leq c\, g(n) \text{ for all } n \geq n_0 \}$$




$O(n \log n)$  $O(n^2)$

$$3n^2 + 5n = \Theta(n^2)$$

$$= O(n^2)$$

$$= O(n^3)$$

$$\neq O(n)$$

$$f(n) \in \Theta(g(n)) \Rightarrow f(n) \in O(g(n))$$

$$\not\Leftarrow$$

$$3n^2 + 5n = \Theta(n^3) \not\Leftarrow 3n^2 + 5n = O(n^3)$$

$$\Omega(g(n)) = \{ f(n) : \text{there exist positive constants } c > 0 \text{ and } n_0 \text{ such that}$$

$$0 \leq c\, g(n) \leq f(n)$$

$$\text{for all } n \geq n_0 \}$$